

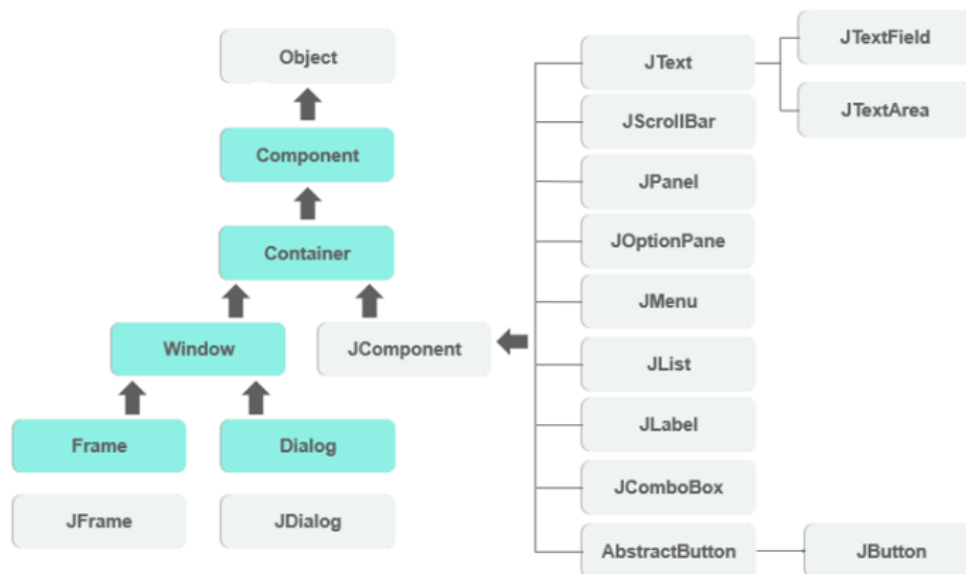
GUI Programming using Java

A **graphical user interface (GUI)** presents a user-friendly mechanism for interacting with an application. A GUI component is an object with which the user interacts via the mouse, the keyboard or another form of input, such as voice recognition. There are various Java GUI components such as **Abstract Window Toolkit (AWT)** in package **java.awt** and **Swing GUI components** from the **javax.swing** package .

AWT	SWING
Platform Dependent	Platform Independent
Does not follow MVC	Follows MVC
Lesser Components	More powerful components
Does not support pluggable look and feel	Supports pluggable look and feel
Heavyweight	Lightweight

There are two groups of classes (in package javax.swing): **containers** and **components**. A container is used to hold components. A container can also hold containers because it is a (subclass of) component.

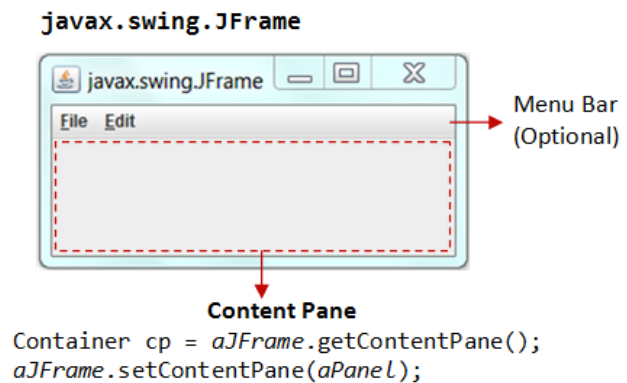
Java Swing Class Hierarchy



Swing applications require a top-level container. There are three top-level containers in Swing:

1. **JFrame**: used for the application's main window (with an icon, a title, minimize/maximize/close buttons, an optional menu-bar, and a content-pane).
2. **JDialog**: used for secondary pop-up window (with a title, a close button, and a content-pane).
3. **JApplet**: used for the applet's display-area (content-pane) inside a browser's window.

There are secondary containers (such as **JPanel**) which can be used to group and layout relevant components.

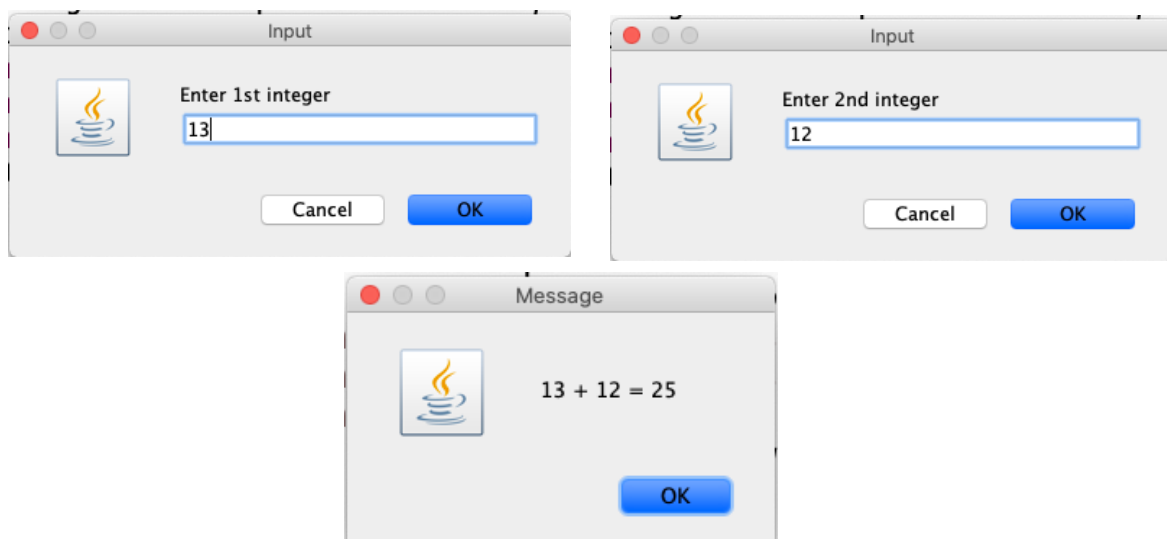


Some basic GUI components

Component	Description
JLabel	Displays uneditable text and/or icons.
TextField	Typically receives input from the user.
Button	Triggers an event when clicked with the mouse.
CheckBox	Specifies an option that can be selected or not selected.
ComboBox	A drop-down list of items from which the user can make a selection.
List	A list of items from which the user can make a selection by clicking on any one of them. Multiple elements can be selected.
JPanel	An area in which components can be placed and organized.

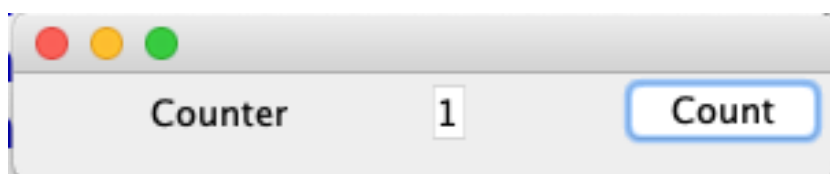
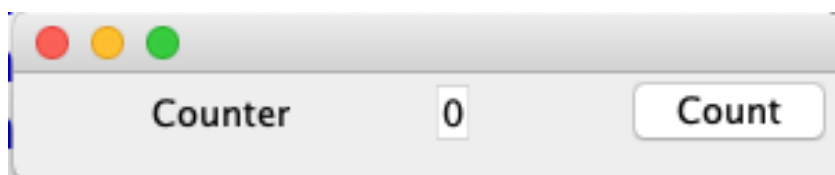
Example 1,

```
3 import javax.swing.JOptionPane;  
4  
5 public class GUIDialog {  
6  
7     public static void main(String[] args) {  
8         String str1 = JOptionPane.showInputDialog("Enter 1st integer");  
9         String str2 = JOptionPane.showInputDialog("Enter 2nd integer");  
10        int num1 = Integer.parseInt(str1);  
11        int num2 = Integer.parseInt(str2);  
12        int sum = num1 + num2;  
13        JOptionPane.showMessageDialog(null, num1 + " + " + num2 + " = " + sum);  
14    }  
15 }  
16 }
```



Example 2,

```
3 import java.awt.FlowLayout;
4 import java.awt.event.ActionEvent;
5 import java.awt.event.ActionListener;
6 import javax.swing.JButton;
7 import javax.swing.JFrame;
8 import javax.swing.JLabel;
9 import javax.swing.JTextField;
10
11 public class SwingCounter extends JFrame {
12     private JTextField tfCount;
13     private JButton btnCount;
14     private int count = 0;
15
16     public SwingCounter() {
17         setLayout(new FlowLayout(FlowLayout.LEFT, 50, 0));
18         add(new JLabel("Counter"));
19         tfCount = new JTextField("0");
20         tfCount.setEditable(false);
21         add(tfCount);
22         btnCount = new JButton("Count");
23         add(btnCount);
24         btnCount.addActionListener(new ButtonClickListener());
25     }
26     public class ButtonClickListener implements ActionListener {
27         public void actionPerformed(ActionEvent arg0){
28             count++; tfCount.setText(count + "");
29         }
30     }
31
32     public static void main(String[] args) {
33         SwingCounter sc = new SwingCounter(); // create Swing Counter
34         sc.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
35         sc.setSize(300, 60); // set frame size
36         sc.setVisible(true); // display frame
37     }
38 }
```



Example 3,

```
12 public class GUIFrame implements ActionListener {
13     private static JLabel userLabel;
14     private static JTextField userText;
15     private static JLabel passwordLabel;
16     private static JPasswordField passwordText;
17     private static JButton button;
18     private static JLabel success;
19
20     public static void main(String[] args) {
21         JFrame frame= new JFrame();
22         JPanel panel= new JPanel();
23         frame.setSize(300, 200);
24         frame.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE );
25         frame.add(panel);
26         panel.setLayout(null);
27
28         userLabel= new JLabel("Username");
29         userLabel.setBounds(10, 20, 80, 25);
30         panel.add(userLabel);
31
32         userText= new JTextField(20);
33         userText.setBounds(100, 20, 160, 25);
34         panel.add(userText);
35
36         passwordLabel= new JLabel("Password");
37         passwordLabel.setBounds(10, 50, 80, 25);
38         panel.add(passwordLabel);
39
40         passwordText= new JPasswordField();
41         passwordText.setBounds(100, 50, 160, 25);
42         panel.add(passwordText);
43
44         button= new JButton("Login");
45         button.setBounds(130, 80, 80, 25);
46         button.addActionListener(new GUIFrame());
47         panel.add(button);
48
49         success= new JLabel();
50         success.setBounds(10, 110, 300, 25);
51         panel.add(success);
52
53         frame.setVisible(true);
54     }
55     @Override
56     public void actionPerformed(ActionEvent e) {
57         String user= userText.getText();
58         String pass= passwordText.getText();
59         if(user.equals("Ali") && pass.equals("1234"))
60             success.setText("Login Successful!");
61         else
62             success.setText("Login Faild");
63     }
64 }
```

